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General Comments : Single HW01 submission document consolidating the Problem Description, the Solution Outline, and
the conceptual Domain Model of the CSE305 Library Management System.
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HW01 — Problem Description, Solution Outline & Domain Model

Library Management System (LMS)

Course	CSE305 Software Engineering — Adnan Menderes University (ADU), Department of Computer Engineering
Instructor	Assoc. Prof. Dr. Fatih Soygazi
Project	Library Management System (LMS)
Team & Scrum roles	**Ibrahim Temur** (`231805033`) — Product Owner · **Omer Faruk Kuş** (`231805026`) — Scrum Master (both: Developer, Architect, QA, Reviewer)
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Archive	`cse305_hw01_231805033_231805026.zip` (created with `tools\package 01`)
GitHub repository	`https://github.com/ibrahimtemur/cse305-library-management-system`

> **Note**
> **How to read this document.** Parts 1 and 2 are concise consolidations. Details live in the companion documents `docs/01-proposal/PROJECT_PROPOSAL.md` and `docs/01-proposal/SOLUTION_OUTLINE.md` (both version `1.2`), and in the Inception documents `docs/02-inception/BUSINESS_RULES.md` (`BR-001` .. `BR-018`), `docs/02-inception/DOMAIN_GLOSSARY.md` , `docs/02-inception/REQUIREMENTS.md` (`FR-001` .. `FR-032`), and `docs/02-inception/USE_CASES.md` (`UC-01` .. `UC-11`). Part 3 (Domain Model) is specified in full here. The seed data is documented in `data/README.md` .

Contents: Part 1 — Problem Description · Part 2 — Solution Outline · Part 3 — Domain Model · Revision History

Part 1 — Problem Description

1.1 Problem Statement

University, faculty, and departmental libraries that still use paper ledgers or ad-hoc spreadsheets lose time and accuracy at the circulation desk:

1. **Manual tracking errors** — checkouts, due dates, renewals, and returns are recorded by hand (mistyped ISBNs, duplicate members, overwritten return dates, books borrowed on someone else's behalf).
2. **Untracked physical copies** — several copies of one title cannot be told apart; copies that are loaned, held for a reservation, under repair, or lost look like shelf stock.
3. **Inconsistent overdue and fine handling** — member-type quotas, overdue days, fines, and borrowing blocks are applied inconsistently.
4. **No real-time visibility** — borrowers cannot check shelf availability, their own due dates and fines, or join a fair waiting queue for a high-demand title.

1.2 Motivation

The LMS is a self-contained, offline-capable web application that automates catalog management, copy-level circulation, FIFO reservation queues, and deterministic fine calculation under the **confirmed lending rules of the ADU Library** (verified on 2026-10-22, see `BUSINESS_RULES.md`). For CSE305 the domain is a realistic vehicle for Agile Scrum, UP-style modeling, GoF/GRASP patterns, UI-first prototyping, and TDD.

1.3 Stakeholders and Roles

Role (`BR-001`)	Who & representative seeded personas (`docs/02-inception/PERSONAS.md`)	Main goals
Librarian / Admin (`ROLE_LIBRARIAN`)	Circulation staff and administrator (one unified role, `ASSUMPTION-01` ; e.g., `Enginiz Yıldırım` `syn-lib-0001`)	Maintain catalog and copies, manage members, run checkout/return/renewal, settle fines, process lost-book petitions, monitor the dashboard (`UC-03` .. `UC-07` , `UC-09` , `UC-11`)
Member (`ROLE_MEMBER`)	`STUDENT` (`Tutkucan Ergül` `SYN-STU-0001`), `ACADEMIC_STAFF` (`Atasagun Öcalan` `SYN-STF-0003`), or `ADMINISTRATIVE_STAFF` (`Doğannur Güçlü` `SYN-ADM-0001`)	Search the catalog, see availability, track own loans/fines, renew, reserve (`UC-01` , `UC-07` , `UC-08` , `UC-10`)
Guest (`ROLE_GUEST`)	Unauthenticated visitor	Read-only catalog search and availability (`UC-01`)
Course instructor / Scrum team	Evaluator (`Assoc. Prof. Dr. Fatih Soygazi`) / PO (`Ibrahim Temur`) and SM (`Omer Faruk Kuş`)	Traceability from vision to tests; sustainable velocity (`docs/02-inception/VISION_AND_CONTEXT.md`)

Confirmed borrowing parameters per member type (`BR-002` , `BR-003`):

Member type	Max simultaneous `ACTIVE` loans	Loan period
`ACADEMIC_STAFF`	7	30 calendar days
`ADMINISTRATIVE_STAFF`	3	30 calendar days
`STUDENT`	3	30 calendar days

Other confirmed rules in force: overdue fine `0.25 TRY` (25 kuruş) per overdue day per loan with no cap and no grace period (`BR-007` .. `BR-009`); no new loan or renewal while a loan is overdue or a fine is unpaid (`BR-011`); one renewal of +30 days if not overdue and not reserved (`BR-004` , `BR-005`); 30-day re-borrowing cooldown after returning a renewed title (`BR-006`); FIFO title-level reservations with a 3-day pickup hold (`BR-012` , `BR-013`); non-circulating reference works (`BR-014`); lost-book petition freezes fines, 30-day replacement window, then twice the appraised value (`BR-015`); personal ID card, no proxy borrowing (`BR-016`).

1.4 Scope

In scope	Out of scope
Bibliographic catalog (books only, including `referenceOnly` works) seeded from Open Library	Periodicals and unpublished theses (`CON-002`)
Copy-level inventory with lifecycle states (`BR-017`)	Live ADU systems: OBIS/OBS, LDAP, SSO, smart-card readers (`CON-003`)
Member management for three member types	Payment gateways, banking APIs, SMTP/SMS (`CON-004`); fines are settled at the desk, hold notices are in-app
Checkout, return, renewal, overdue detection, fines, waivers, lost-book handling	RFID gates and scanner drivers (keyboard-emulated barcode input is supported)
FIFO reservation queue with pickup hold	E-book hosting and DRM
Role-based server-side web UI (Spring MVC + Thymeleaf), designed first in Stitch MCP	Multi-branch inter-library shipping

1.5 Prioritized Features (MoSCoW)

Priority	ID	Feature	`FR-*`	`UC-*`
Must	`M1`	Catalog & book copy management	`FR-001` .. `FR-005`	`UC-01` , `UC-03`
Must	`M2`	Member management (3 member types)	`FR-006` .. `FR-008`	`UC-04`
Must	`M3`	Loan checkout & return	`FR-009` .. `FR-013`	`UC-05` , `UC-06`
Must	`M4`	Overdue detection & fine calculation	`FR-014` , `FR-015`	`UC-06` , `UC-09` , `UC-11`
Must	`M5`	Authentication, role-based views, self-service	`FR-016` .. `FR-018`	`UC-02` , `UC-10`
Must	`M6`	Reproducible seed data pipeline & startup loader	`FR-019`	startup
Should	`S1`	Reservation (hold) queue & pickup hold	`FR-020` .. `FR-023`	`UC-08`
Should	`S2`	Loan renewal	`FR-024`	`UC-07`
Should	`S3`	Fine settlement, waiver, lost-copy handling	`FR-025` .. `FR-027`	`UC-09`

Priority	ID	Feature	FR-*	UC-*
Should	S4	Librarian dashboard & overdue report	FR-028, FR-029	UC-11
Could	C1 .. C3	Guest portal; live ISBN lookup; CSV export	FR-030 .. FR-032	UC-01, UC-03, UC-11
Won't	W1	Online payments, external messaging, inter-library shipping	—	—

1.6 Domain Description

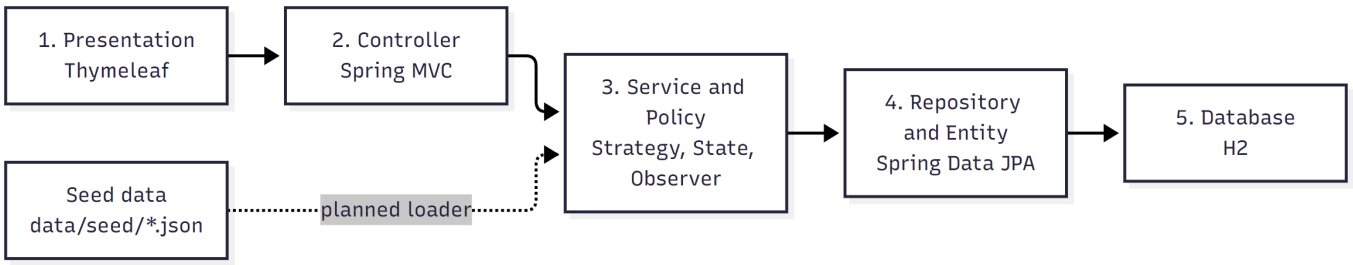
The conceptual domain model in Part 3 is derived directly from the narrative descriptions below. Each sentence group is tagged (**D1**..**D8**) to establish formal traceability from domain text to conceptual classes, attributes, associations, and enumerations (see §3.9):

- **D1.** The library keeps a catalog of **Books**. A Book is a bibliographic title identified by its `isbn13`. It has a `title`, an optional `subtitle`, one or more `authors`, a `publisher`, a `firstPublishYear`, a `category`, a list of `subjects`, a `pageCount`, a cover reference (`coverId`, `coverUrl`), and an `openLibraryWorkKey`. Some Books are `referenceOnly` (encyclopedias, dictionaries, catalogs): they stay in the library and are never lent.
- **D2.** A Book *has* zero or more **Book Copies**. A Book Copy is one physical item of exactly one Book, identified by a unique `barcode`. Each Book Copy has a `status`: AVAILABLE, LOANED, RESERVED, MAINTENANCE or LOST. A Book must be cataloged before copies of it are added.
- **D3.** The library serves **Members**. Each Member has a `memberId` (which stands for the personal ID card), a `firstName`, a `lastName`, an `email`, a `department` and a `registeredOn` date. A Member has a `memberType` (STUDENT, ACADEMIC_STAFF or ADMINISTRATIVE_STAFF) and a membership `status` (ACTIVE or SUSPENDED). An academic staff member may hold at most seven active loans, a student or an administrative staff member at most three.
- **D4. Librarians** operate the circulation desk. Each Librarian has a `username`, a `displayName`, an `email` and a system `role` (`ROLE_LIBRARIAN`).
- **D5.** A Member *borrows* a Book Copy, and every borrowing is recorded as a **Loan** identified by a unique `loanId`. A Loan has a `checkoutDate`, a `dueDate` (30 days after checkout, 30 days later again if renewed once), a `returnDate`, a `renewalCount`, a `renewedOn` date, a `lostReportedOn` date and a `status` (ACTIVE, RETURNED or LOST). A Member has zero or more Loans, each Loan belongs to exactly one Member and concerns exactly one Book Copy. A Book Copy is *lent in* zero or more Loans over time, but in at most one active Loan at any moment.
- **D6.** When a Loan is overdue, the library charges a **Fine** for it identified by a unique `fineId`. A Fine has an `overdueDays` count, an `amount` (0.25 TRY per overdue day, with no maximum), a `currency`, a `status` (UNPAID, PAID or WAIVED), an `assessedOn` date and a `settledOn` date. A Loan *incurs* at most one Fine, and each Fine belongs to exactly one Loan. A Librarian *settles* a Fine by recording the payment or by waiving it; a Librarian settles zero or more Fines, and a settled Fine is settled by at most one Librarian (an unpaid fine is not settled yet).
- **D7.** When no copy of a Book is available, a Member may place a **Reservation** on the Book. A Reservation has a `reservationId`, a `queuePosition` in the first-come-first-served queue of that Book, a `status` (WAITING or READY) and a `createdOn` date. A Member *places* zero or more Reservations and a Book *is reserved through* zero or more Reservations. When a copy is returned, the first waiting Reservation becomes READY and *holds* exactly one Book Copy until its `holdUntil` date, three days after its `readyOn` date. A Book Copy is held by at most one Reservation.
- **D8.** If a Member loses a Book Copy, the Member reports it on a `lostReportedOn` date, the Loan becomes LOST, the fine for that Loan stops growing on that date, and the Book Copy becomes LOST. The Valuation Commission's `appraisedValue` and the resulting compensation of twice that value are entered by the Librarian when the book is not replaced within 30 days, deriving the `compensationAmount` (`2.00 * appraisedValue`).

Part 2 — Solution Outline

2.1 Architecture

A five-tier layered web architecture (`NFR-010`): **Thymeleaf UI** → **Controller** → **Service/Policy** → **Repository/Entity** → **H2**. Controllers only coordinate HTTP requests (GRASP Controller); transactional services and policy/pattern classes hold the business rules (`NFR-003`); repositories map the persistent domain concepts to H2 (file mode at runtime, in-memory mode in tests). Details and the full diagram: `docs/01-proposal/SOLUTION_OUTLINE.md` §1.



2.2 Technology Stack

Area	Choice	Status
Language / framework	Java 17 target (built on JDK 25 LTS), Spring Boot `3.5.16`	in `pom.xml`
Web / persistence	Spring MVC + Thymeleaf, Spring Data JPA / Hibernate, Jakarta Bean Validation, H2	in `pom.xml`
Security	Spring Security with BCrypt (`FR-016` , `FR-017` , `NFR-007` , `NFR-008`)	**planned**, not yet in `pom.xml`
Build / tests	Maven `3.9.16` + wrapper (`.\mvnw.cmd`), JUnit 5, Mockito, Spring Boot Test / MockMvc	working
Seed tooling	Python 3.11 scripts in `tools/data/` (Open Library fetcher, `Faker` `tr_TR`), 80 offline `unittest` checks	working
UI design	Stitch MCP	working

2.3 Data Strategy

Real bibliographic metadata plus 100% synthetic operational data, versioned under `data/seed/` so that startup and tests run offline (`NFR-004` , `NFR-009` , `RISK-03`). Seed v2 (reference date `2026-10-22`) contains:

File	Rows	Content
`books-openlibrary.json`	782	Open Library books, 3 of them `referenceOnly`
`members.json`	60	40 `STUDENT` , 10 `ACADEMIC_STAFF` , 10 `ADMINISTRATIVE_STAFF`
`librarians.json`	3	`ROLE_LIBRARIAN` accounts, no credentials
`copies.json`	1,706	1 to 4 copies per title
`loans.json`	164	59 `ACTIVE` , 100 `RETURNED` , 5 `LOST`
`fines.json`	47	26 `UNPAID` , 16 `PAID` , 5 `WAIVED`
`reservations.json`	22	19 `WAITING` , 3 `READY` , across 12 titles

Conventions: members `SYN-STU-####` , `SYN-STF-####` , `SYN-ADM-####` ; librarians `syn-lib-####` ; e-mail domain `synthetic.lms.local` ; barcodes `LMS-COPY-####` ; UTF-8, LF, JSON arrays, ISO-8601 dates, natural-key references, fine `amount` as a two-decimal string. No passwords, real personal data, or book prices exist in the seed files (`BR-018` , `ASSUMPTION-15`); demo credentials are BCrypt-hashed at startup (`ASSUMPTION-16`). Full format description: `data/README.md` .

2.4 UI-First Workflow

Requirements and personas → screen design in **Stitch MCP** → HTML prototype and screenshot saved under `docs/03-design/ui/` → Thymeleaf template and Spring MVC controller → MockMvc and browser verification (`AGENTS.md` §6, `NFR-001`). Eight core screens are planned (login, catalog and book detail, circulation desk, catalog/copy management, member management, self-service portal, fines and lost-book desk, dashboard); see `docs/01-proposal/SOLUTION_OUTLINE.md` §3.

2.5 Design Patterns (design level, elaborated in HW03)

Pattern	Where	Why
Strategy	`LoanPolicyStrategy` , `FineCalculationStrategy`	Quotas (7/3/3), loan period, and fine tariff are policy parameters that may change (protected variations); simplified if HW03 shows a parameter table suffices
State	`BookCopyState`	Strict copy lifecycle (`BR-017` , `FR-005`)

Pattern	Where	Why
Observer	`BookReturnedEvent` and listeners	Return triggers hold promotion and notification without coupling services (`FR-022`)
Factory / Builder	`LoanFactory`, test builders	Valid aggregates by construction
Adapter / Facade	`OpenLibraryCatalogAdapter`	Keeps third-party JSON out of entities and controllers
GRASP	Information Expert, Creator, Controller, Low Coupling, High Cohesion, Protected Variations	Responsibility assignment

None of these design-level classes appear in the Domain Model of Part 3.

2.6 Agentic Development Approach

AI agents draft documents, prototypes, and test/code scaffolding under the five engineering hats and the `Plan → Execute → Record` cadence of `AGENTS.md`; humans own scope, priorities, business rules, acceptance, and every merge into `main`. Quality gates: plan approval, automated tests, a cross-model reviewer pass, and honest audit logging in `docs/process/ai-usage-log.md` (`Pending human review` unless a named person confirmed the review). Details: `docs/01-proposal/PROJECT_PROPOSAL.md` §6.3.

2.7 Test Strategy

Level	Tooling	Scope
Domain and pattern unit tests	JUnit 5, no Spring context	Quotas 3 and 7, exact due date (0 overdue days) versus next day (1), `0.25 TRY/day` without cap, 30-day cooldown boundary, fine freeze on `lostReportedOn`, copy state transitions
Repository slices	`@DataJpaTest`, in-memory H2	Mappings and unique keys (`isbn13`, `barcode`, `memberId`, `username`)
Service / web integration	`@SpringBootTest`, `@WebMvcTest`, MockMvc	Checkout, return with hold promotion, blocks, role-based access
Seed data	Python `unittest` (80 checks)	Counts, foreign keys, rule invariants, reproducibility

Gate: `.\mvnw.cmd -q test` and `tools\data\.venv\Scripts\python.exe -m unittest discover -s tools\data\tests` must pass before every merge.

Part 3 — Domain Model

The domain model follows Larman, *Applying UML and Patterns* (3rd Ed., domain model chapter): it shows **conceptual classes** of the problem domain with attributes, named associations, and multiplicities. It deliberately contains **no** controllers, services, repositories, DTOs, strategies, or state classes — those are design concepts (HW03). Attribute names are the field names of `data/seed/*.json` and `docs/02-inception/DOMAIN_GLOSSARY.md`. **Foreign-key fields** (`isbn13` in a copy or reservation, `memberId`, `copyBarcode`, `loanId`, `readyCopyBarcode`, `settledBy`) are **not** shown as attributes; the corresponding **associations** express them (see the association table). The model was derived directly from the domain description narrative in §1.6 (see §3.9 for the complete text-to-model traceability mapping).

3.1 Class Diagram

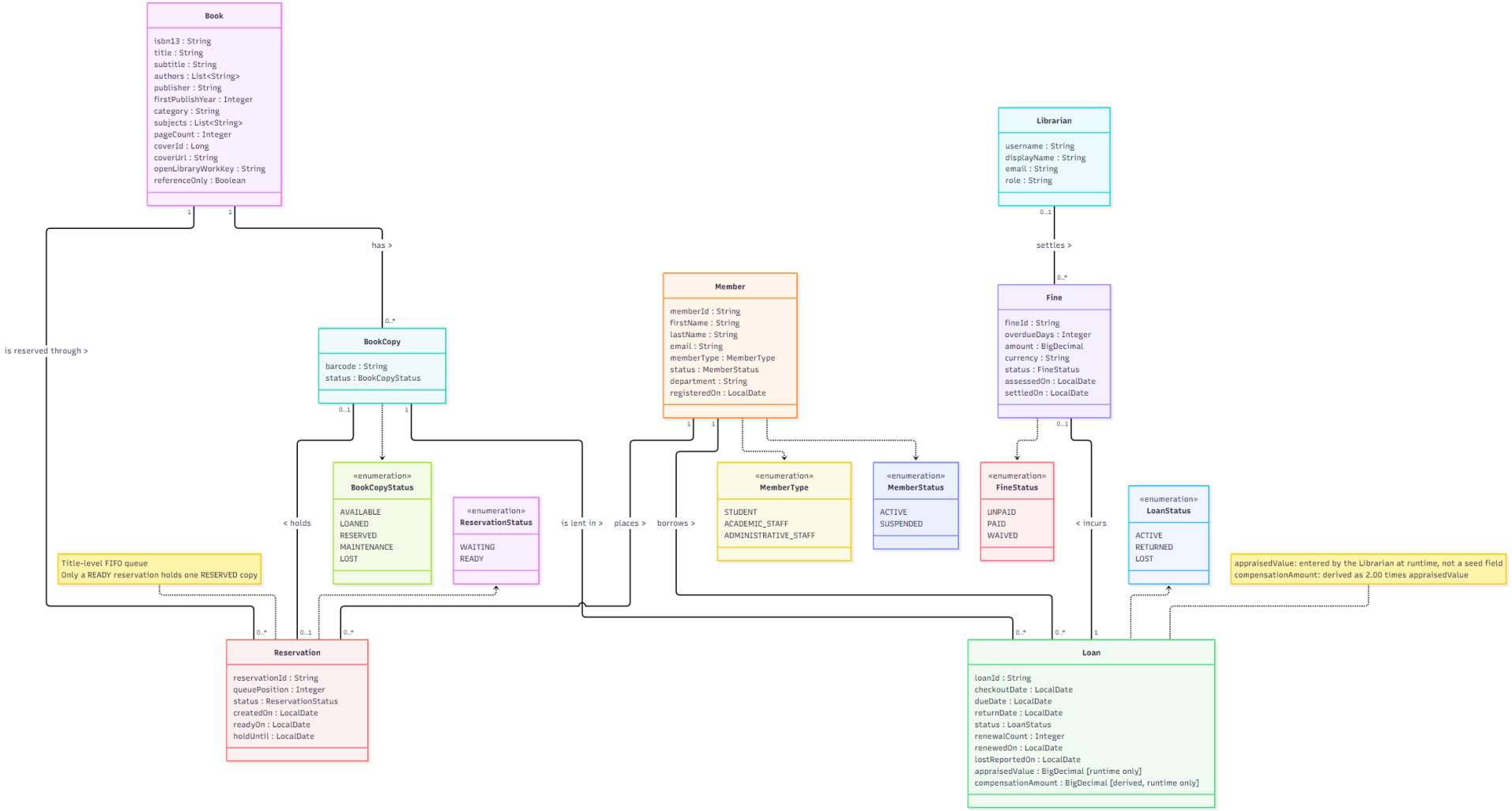


Figure 1 — Conceptual class diagram including the seven conceptual classes and six domain enumerations (vector version: `docs/01-proposal/images/domain-model.svg` ; source: `docs/01-proposal/images/domain-model.mmd`). Attribute types follow the seed formats; the six enumeration types are also summarized with seed distributions in §3.2.

▶ Mermaid source (identical to `images/domain-model.mmd`)

3.2 Enumerations

Enumeration	Used by	Values	Seed distribution
`MemberType`	`Member.memberType`	`STUDENT`, `ACADEMIC_STAFF`, `ADMINISTRATIVE_STAFF`	40 / 10 / 10
`MemberStatus`	`Member.status`	`ACTIVE`, `SUSPENDED`	55 / 5
`BookCopyStatus`	`BookCopy.status`	`AVAILABLE`, `LOANED`, `RESERVED`, `MAINTENANCE`, `LOST`	1,624 / 59 / 3 / 15 / 5
`LoanStatus`	`Loan.status`	`ACTIVE`, `RETURNED`, `LOST`	59 / 100 / 5
`FineStatus`	`Fine.status`	`UNPAID`, `PAID`, `WAIVED`	26 / 16 / 5
`ReservationStatus`	`Reservation.status`	`WAITING`, `READY`	19 / 3

`Librarian.role` is the constant string `ROLE_LIBRARIAN` (a single value, hence not an enumeration). `Guest` and the `ROLE_*` authorities are actors/roles of the system, not persistent conceptual classes (`BR-001`).

Domain parameters by member type. `memberType` is an enumeration, not a class hierarchy, because members differ only in rule parameters, not in attributes or associations (Larman: subclass only when attributes, associations, or behavior differ). The parameters are values of rules `BR-002` and `BR-003`, not model elements:

`MemberType`	Max `ACTIVE` loans	Loan period
`ACADEMIC_STAFF`	7	30 days
`ADMINISTRATIVE_STAFF`	3	30 days
`STUDENT`	3	30 days

3.3 Conceptual Classes, Seed Files, and Traceability

Conceptual class	Meaning in the domain	Seed file	Related `FR-*`	Related `BR-*`	Related `UC-*`
`Book`	A bibliographic title identified by a checksum-valid `isbn13`; may be `referenceOnly` (non-circulating)	`books-openlibrary.json`	`FR-001`, `FR-002`, `FR-003`, `FR-031`	`BR-014`, `BR-017`	`UC-01`, `UC-03`
`BookCopy`	One physical item of a title, identified by its `barcode`; has a lifecycle `status`	`copies.json`	`FR-003`, `FR-004`, `FR-005`, `FR-010`, `FR-012`	`BR-013`, `BR-014`, `BR-017`	`UC-03`, `UC-05`, `UC-06`
`Member`	A registered borrower (student, academic staff, or administrative staff)	`members.json`	`FR-006`, `FR-007`, `FR-008`, `FR-009`	`BR-002`, `BR-003`, `BR-006`, `BR-011`, `BR-016`, `BR-018`	`UC-04`, `UC-05`, `UC-10`
`Librarian`	Library staff member who operates the desk and settles fines (unified with the administrator, `ASSUMPTION-01`)	`librarians.json`	`FR-016`, `FR-017`, `FR-025`, `FR-026`	`BR-001`, `BR-010`, `BR-018`	`UC-02`, `UC-05`, `UC-06`, `UC-09`
`Loan`	The lending of one `BookCopy` to one `Member`, from checkout until return or loss	`loans.json`	`FR-009`..`FR-014`, `FR-024`, `FR-027`	`BR-003`..`BR-007`, `BR-011`, `BR-013`, `BR-015`, `BR-016`, `BR-017`	`UC-05`, `UC-06`, `UC-07`, `UC-09`
`Fine`	The monetary overdue penalty of one loan, with its settlement (`PAID` or `WAIVED`)	`fines.json`	`FR-014`, `FR-015`, `FR-025`, `FR-026`, `FR-029`	`BR-007`..`BR-011`, `BR-015`	`UC-06`, `UC-09`, `UC-11`
`Reservation`	A title-level hold in a FIFO queue; when `READY` it holds one `RESERVED` copy until `holdUntil`	`reservations.json`	`FR-020`, `FR-021`, `FR-022`, `FR-023`	`BR-012`, `BR-013`, `BR-014`	`UC-06`, `UC-08`, `UC-10`

3.4 Attribute Dictionary

`?` marks attributes that may be `null` in the seed. Types follow `data/README.md` §6.

Class	Attribute	Type	Meaning
`Book`	`isbn13`	String	Unique 13-digit ISBN with valid check digit
	`title`, `subtitle?`	String	Title; optional subtitle

Class	Attribute	Type	Meaning
	`authors`	List of String	1 to 5 author names
	`publisher`	String	Publisher of the selected edition
	`firstPublishYear`	Integer	First publication year of the work
	`category`	String	One of 11 catalog categories
	`subjects`	List of String	Up to 5 raw Open Library subjects
	`pageCount`	Integer	Work-level median page count (may differ from the specific edition)
	`coverId`, `coverUrl`	Long, String	Open Library cover reference (image not stored)
	`openLibraryWorkKey`	String	Open Library work key
	`referenceOnly`	Boolean	`true` for non-circulating encyclopedias, dictionaries, catalogs (`BR-014`)
`BookCopy`	`barcode`	String	Unique `LMS-COPY-####`
	`status`	BookCopyStatus	Lifecycle state
`Member`	`memberId`	String	Unique `SYN-STU-####`, `SYN-STF-####`, or `SYN-ADM-####`; stands for the personal ID card (`BR-016`)
	`firstName`, `lastName`	String	Synthetic name
	`email`	String	`*@synthetic.lms.local`
	`memberType`, `status`	MemberType, MemberStatus	Member type and membership state
	`department`	String	Generic fictional department label
	`registeredOn`	LocalDate	Registration date
`Librarian`	`username`	String	Unique `syn-lib-####`
	`displayName`, `email`	String	Synthetic name and e-mail
	`role`	String	Constant `ROLE_LIBRARIAN`; no credential field in the seed
`Loan`	`loanId`	String	Unique `LOAN-####`
	`checkoutDate`, `dueDate`	LocalDate	`dueDate = checkoutDate + 30 * (1 + renewalCount)` days
	`returnDate?`	LocalDate	Set only for `RETURNED` loans
	`status`	LoanStatus	`ACTIVE`, `RETURNED`, or `LOST`
	`renewalCount`	Integer	`0` or `1`
	`renewedOn?`	LocalDate	Date of the renewal
	`lostReportedOn?`	LocalDate	Lost-book petition date; freezes fine accrual (`BR-015`)
	`appraisedValue` *(runtime only)*	BigDecimal	Valuation Commission value entered by the Librarian; **not in the seed** (`ASSUMPTION-15`)
	`compensationAmount` *(derived, runtime only)*	BigDecimal	`2.00 * appraisedValue`; **not in the seed**
`Fine`	`fineId`	String	Unique `FINE-####`
	`overdueDays`	Integer	Overdue days up to the effective end date (`BR-007`)
	`amount`, `currency`	BigDecimal, String	`0.25 * overdueDays`, no cap; `TRY`
	`status`	FineStatus	`UNPAID`, `PAID`, or `WAIVED`
	`assessedOn`	LocalDate	Return date, lost-petition date, or evaluation date
	`settledOn?`	LocalDate	Settlement date; `null` while `UNPAID`
`Reservation`	`reservationId`	String	Unique `RES-####`
	`queuePosition`	Integer	FIFO position `1..n` within the title's queue
	`status`	ReservationStatus	`WAITING` or `READY`
	`createdOn`	LocalDate	Placement date (FIFO order)

Class	Attribute	Type	Meaning
	`readyOn?`, `holdUntil?`	LocalDate	Hold start and `readyOn + 3 days`; only for `READY`

3.5 Associations and Multiplicities

#	Association (read direction)	Multiplicity	Seed foreign key	Rule / invariant
A1	`Book` has `BookCopy`	1 — `0..*`	`copies.isbn13`	A title is cataloged before copies are added (`FR-001`, `FR-004`); seed has 1 to 4 per title (`ASSUMPTION-09`, `BR-017`)
A2	`Member` borrows `Loan`	1 — `0..*`	`loans.memberId`	Active-loan quota per member type (`BR-002`); blocks (`BR-011`)
A3	`BookCopy` is lent in `Loan`	1 — `0..*`	`loans.copyBarcode`	At most one `ACTIVE` loan per copy at any time (`BR-017`)
A4	`Loan` incurs `Fine`	1 — `0..1`	`fines.loanId` (unique)	At most one overdue fine per loan (`BR-010`); every fine belongs to exactly one loan
A5	`Librarian` settles `Fine`	`0..1` — `0..*`	`fines.settledBy`	`null` while `UNPAID`; set when `PAID` or `WAIVED` (`BR-010`)
A6	`Member` places `Reservation`	1 — `0..*`	`reservations.memberId`	Eligibility (`BR-012`)
A7	`Book` is reserved through `Reservation`	1 — `0..*`	`reservations.isbn13`	Title-level FIFO queue (`BR-012`); never on a `referenceOnly` book (`BR-014`)
A8	`Reservation` holds `BookCopy`	`0..1` — `0..1`	`reservations.readyCopyBarcode`	Only a `READY` reservation holds a copy, and a `RESERVED` copy is held by exactly one `READY` reservation (`BR-013`, `BR-017`)

Derived association (not drawn). `fines.memberId` duplicates `Loan.member` (verified for all 47 seed fines). Larman advises against redundant associations, so a `Member`'s fines are reached through `Member → Loan → Fine`.

Cross-association invariants (enforced in design and tests, shown here for completeness): `BookCopy.status = LOANED` ⇔ the copy has an `ACTIVE` loan; `BookCopy.status = LOST` ⇔ the copy has a `LOST` loan (`returnDate = null`, `lostReportedOn ≠ null`); `BookCopy.status = RESERVED` ⇔ a `READY` reservation holds it; all copies of a `referenceOnly` book stay `AVAILABLE` (`BR-014`, `BR-017`).

3.6 Key Modelling Decisions

1. **`Book` versus `BookCopy`.** A title (`isbn13`) and a physical item (`barcode`) have different identities, attributes, and lifecycles: availability, loans, loss, and maintenance concern copies; catalog data, reference status, and reservations concern the title. Merging them would make multi-copy lending and per-copy state impossible.
2. **`Member` with an enumerated `memberType`, no subclasses.** The three member types differ only in rule parameters (`BR-002`, `BR-003`), so an enumeration is enough; the parameters belong to the rules, not to the model.
3. **`Librarian` is a separate class.** Librarians and members are distinct people with different data (`librarians.json` versus `members.json`), roles, and responsibilities; a librarian appears in the domain as the person who settles a `Fine`.
4. **`Fine` is its own concept.** A fine has its own lifecycle (`UNPAID` → `PAID` or `WAIVED`), amount, assessment and settlement dates, and an accountable librarian; at most one overdue fine exists per loan (`BR-010`), and it blocks borrowing while unpaid (`BR-011`). Folding it into `Loan` would hide this lifecycle.
5. **Title-level FIFO reservations.** A `Reservation` is placed on a `Book`, not on a copy (`BR-012`); `queuePosition` and `createdOn` give the FIFO order. Only when a copy is returned does a `READY` reservation hold one specific `BookCopy` (`readyCopyBarcode`, `BR-013`).
6. **Why a lost loan has its own status.** `LOST` is a closed outcome that differs from `RETURNED`: the copy never came back (`returnDate = null`), fine accrual stopped on `lostReportedOn`, and the copy becomes `LOST`. Overloading `RETURNED` would falsely free the copy, break the invariant "`LOST` copy ⇔ `LOST` loan", and lose the fine-freeze date (`BR-015`, `BR-017`).
7. **Why `appraisedValue` is runtime-only.** Open Library data contains no prices, and inventing fake prices in the seed would be false data. The Valuation Commission's value is entered by the Librarian when a lost copy is not replaced within 30 days (`ASSUMPTION-15`); `compensationAmount` is derived as `2.00 * appraisedValue`. Both appear on `Loan` as attributes marked *runtime only* and are not seed fields.
8. **Associations instead of foreign keys.** Following Larman, relationships are drawn as named associations with multiplicities; the seed's natural-key fields are the implementation of those associations and are mapped in §3.5. The redundant `Fine`–`Member` link is omitted.
9. **`Reservation` states `WAITING` and `READY` only.** This equals the glossary and the seed. A reservation that is fulfilled, cancelled, or expired leaves the active queue (see Known Limitations).

3.7 Known Limitations

- **Lost-book compensation tracking is deferred to HW03.** The model records `appraisedValue` and the derived `compensationAmount` on `Loan` only. How the compensation charge is **tracked, paid, or waived** (for example its own status, `settledOn`, and `settledBy`, or an extension of `Fine`) is not modelled in HW01 and will be decided in the HW03 design documents. Until then, `FR-025` / `FR-026` (settlement and waiver) are fully modelled for overdue `Fine` records only.
- **Reservation terminal states and history are deferred to HW03.** `ReservationStatus` contains only `WAITING` and `READY`. `FR-011` (fulfilled by checkout) and `FR-023` (cancellation, expiry) remove a reservation from the active queue; whether terminal states such as fulfilled, cancelled, or expired, and a retained reservation history, are needed will be decided in HW03. The seed contains no such records.
- The model is conceptual: it carries no operations, visibility, or persistence mapping.
- Seed dates are relative to `REFERENCE_DATE = 2026-10-22`; shifting them at startup is open item `OPEN-04`.

3.8 Concepts Intentionally Left to the HW03 Design Documents

Design-level concept	Why it is not in the domain model
`LoanPolicyStrategy`, `FineCalculationStrategy` and their implementations	Strategy classes are design solutions for policy variation, not domain concepts
`BookCopyState` and its state classes	State pattern realizes the `BookCopyStatus` lifecycle
`BookReturnedEvent`, `LoanOverdueEvent`, listeners	Observer mechanism
`LoanFactory`, builders	Creational design
`Notification` (in-app hold-ready message, `BR-013`)	Runtime artifact; its need as a persistent concept is a design decision
User account / credentials / security roles	Login data is created at startup (`ASSUMPTION-16`); not in the seed
Controllers, services, repositories, DTOs, JPA entity mappings	Layered-architecture elements
Compensation charge tracking; reservation history	See Known Limitations

3.9 From the Domain Description to the Model

The domain model in this chapter is directly grounded in the domain narrative of §1.6. The table below traces each narrative paragraph (**D1**..**D8**) to its corresponding conceptual classes, attributes, associations, and enumerations in the domain model. All 7 conceptual classes, 50 attributes, 6 enumerations (18 values), and 8 associations (A1..A8) are verified with bidirectional traceability.

Paragraph ID	Narrative Concept (Text Name)	Target Model Class	Attributes Captured	Associations Captured	Enumerations & Values Captured	Domain Notes & Context
D1	Book (`Books`)	`Book`	`isbn13`, `title`, `subtitle`, `authors`, `publisher`, `firstPublishYear`, `category`, `subjects`, `pageCount`, `coverId`, `coverUrl`, `openLibraryWorkKey`, `referenceOnly` (13)	—	—	`referenceOnly = true` marks non-circulating reference works (encyclopedias, dictionaries, catalogs; `BR-014`).
D2	Book Copy (`Book Copies`)	`BookCopy`	`barcode`, `status` (2)	A1: `Book` "1" -- "0..*" `BookCopy` : has >	`BookCopyStatus`: `AVAILABLE`, `LOANED`, `RESERVED`, `MAINTENANCE`, `LOST` (5)	A title must be cataloged before copies are added (`BR-017`).
D3	Member (`Members`)	`Member`	`memberId`, `firstName`, `lastName`, `email`, `memberType`, `status`, `department`, `registeredOn` (8)	—	`MemberType`: `STUDENT`, `ACADEMIC_STAFF`, `ADMINISTRATIVE_STAFF` (3) `MemberStatus`: `ACTIVE`, `SUSPENDED` (2)	`memberId` represents the personal ID card (`BR-016`). Loan limits (7 for staff, 3 for students) are rule parameters (`BR-002`).
D4	Librarian (`Librarians`)	`Librarian`	`username`, `displayName`, `email`, `role` (4)	—	`role = "ROLE_LIBRARIAN"` (constant)	Circulation desk is the physical operation point (`ASSUMPTION-01`).

Paragraph ID	Narrative Concept (Text Name)	Target Model Class	Attributes Captured	Associations Captured	Enumerations & Values Captured	Domain Notes & Context
D5	Loan (`Loan`)	`Loan`	`loanId` , `checkoutDate` , `dueDate` , `returnDate` , `status` , `renewalCount` , `renewedOn` , `lostReportedOn` (8 of 10)	A2: `Member` "1" -- "0.." *`Loan`* : *borrows* > *A3:* *`BookCopy`* "*1*" -- "*0..*" `Loan` : is lent in >	`LoanStatus` : `ACTIVE` , `RETURNED` , `LOST` (3)	Standard period 30 days (`BR-003`), max 1 renewal (`BR-005`). At most one active loan per copy (`BR-017`).
D6	Fine (`Fine`)	`Fine`	`fineId` , `overdueDays` , `amount` , `currency` , `status` , `assessedOn` , `settledOn` (7)	A4: `Loan` "1" -- "0..1" `Fine` : < incurs A5: `Librarian` "0..1" -- "0..*" `Fine` : settles >	`FineStatus` : `UNPAID` , `PAID` , `WAIVED` (3)	Fine rate is 0.25 TRY/day without cap (`BR-008` , `BR-009`). Unpaid fine blocks new checkouts (`BR-011`).
D7	Reservation (`Reservation`)	`Reservation`	`reservationId` , `queuePosition` , `status` , `createdOn` , `readyOn` , `holdUntil` (6)	A6: `Member` "1" -- "0.." *`Reservation`* *: places* > *A7:* *`Book`* "*1*" -- "*0..*" `Reservation` : is reserved through > A8: `Reservation` "0..1" -- "0..1" `BookCopy` : holds >	`ReservationStatus` : `WAITING` , `READY` (2)	Title-level FIFO queue (`BR-012`). Pickup hold window is 3 days (`BR-013`).
D8	Lost Loan & Valuation	`Loan` , `BookCopy`	`Loan.lostReportedOn` , `Loan.appraisedValue` , `Loan.compensationAmount` (+2 on `Loan` , completing 10)	—	`LoanStatus.LOST` , `BookCopyStatus.LOST`	Valuation Commission is an institutional body (`BR-015`). `appraisedValue` is a runtime Librarian input; `compensationAmount` is derived (`2.00 * appraisedValue`).

Revision History

Version	Date	Summary
`1.2`	2026-11-01	First consolidated HW01 submission document: Problem Description, Solution Outline, and Domain Model; aligned with confirmed ADU rules and seed v2; schedule updated (HW01 due 2026-11-03); domain description (section 1.6) and text-to-model traceability (section 3.9) added; document date 2026-11-01.